

THERMODYNAMICS

Heat

Movement.

(Movement of heat energy from one place to another/ from one form to another).

Definition → Thermodynamics is a branch of science which deals with transfer of energy and its effect on the physical properties of a substance.

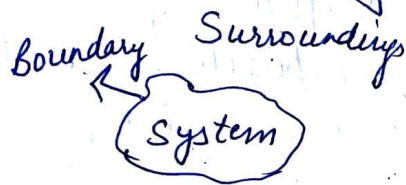
Thermodynamic System ^{Terms}

System → A system is a finite quantity of matter or a region in space chosen for study.

Surroundings → Everything external to the system is called the surroundings or the environment.

Boundary → The separation b/w system and surroundings is called boundary (real or imaginary surface that separates the two). The boundary can be fixed or moving → eg. when a system containing a gas is compressed or expanded.

Universe = System + Surroundings



3 ^{or types} classes of system

(i) Closed system

(ii) Open system

(iii) Isolated system

(i) Closed system → System of fixed mass. Transfer of energy (into or out of the system) can take place, but there is no transfer of mass across the system. e.g. a certain quantity of fluid in a cylinder bounded by a piston.

Open System $\hat{=}$ System in which matter crosses the boundary of the system. There may be energy transfer too. e.g. fluid in a cylinder without a lid/piston.
 $\Rightarrow$ Mass and energy both ~~are~~ can be transferred b/w system and surroundings.

Isolated System $\hat{=}$ No interaction b/w system and the surroundings. Cannot exchange energy or mass $\Rightarrow$ It is of fixed mass and energy. Example $\rightarrow$ Universe.
 (Thermus $\rightarrow$ Not accu etc.)

<u>System</u>	<u>Mass Transfer</u>	<u>Energy transfer.</u>	<u>Example</u>	<u>Boundary</u>
Closed	X	✓	Piston cylinder w/o valves.	
Open	✓	✓	Turbine, compressor, pump etc.	
Isolated	X	X	Universe	

Homogeneous and Heterogeneous Systems

A quantity of matter homogenous throughout in chemical composition and physical structure is called a phase $\rightarrow$ like solid, liquid, gas.

A system consisting of a single phase is homogenous.

A " " of more than one phase is heterogeneous.

e.g. air + water vapour $\rightarrow$ homo ; water + steam $\rightarrow$ hetero.

Pure Substance $\hat{=}$ A system which is Examples:-

- homogenous in composition

- homogenous in chemical aggregation

Liquid water, liquid water + steam,
 mixture of ice + water,
 liquid air + gaseous air NOT a pure substance

Properties of a System.

Every system has certain characteristics by which its physical condition may be described e.g. volume, temp., pressure etc. Such characteristics are called properties of a system.

Properties are of 2 types:

- (i) Extensive properties :- These properties depend on the mass of the system (amount of substances present in the system). e.g. Volume, ~~Mass~~^{Mass}, Energy (except heat and work)
- (ii) Intensive properties :- These properties do not depend on the mass of the system. e.g. temperature, pressure, density.

Extensive $\rightarrow$ Additive (value for an overall system is the sum of its values for the parts into which system is divided). If mass is increased, value of extensive properties also increase.

Intensive $\rightarrow$ Not additive. Value won't change with increase/decrease in mass/size.

e.g. Mass of whole system = Sum of masses of various parts.

Temp. of whole system $\neq$ Sum of temp. of constituent parts.

Extensive properties per unit mass e.g. specific volume (Vol. divided by mass V/m) are called specific properties (lly specific energy) and hence become intensive properties.

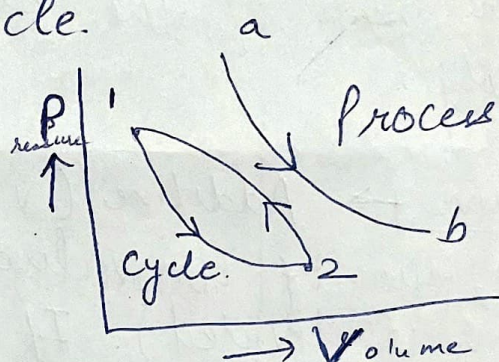
Ques :- Is density m/V intensive or extensive? $\Rightarrow$ Intensive

State of a System $\div$ The condition of system is called state of a system.

- At a given (definite) state, all the properties of a system are fixed i.e. they have definite values.
- A change of state (any operation in which one or more properties of a system changes) is called a process. The series of states through which a system passes during a process is called the path of process.

Thermodynamic Cycle $\div$ A series of state changes such that the final state is identical to the initial state, the system is said to have undergone a thermodynamic cycle.

- Minimum number of processes required for a cycle is 2.



The change in value of any property for a cyclic process is zero.

Thermodynamic Equilibrium $\div$

The thermodynamic state of a system is characterised by its properties. Thermodynamics deals with equilibrium states. [Equilibrium means balance]

In an equilibrium state, there are no unbalanced forces within the system. In other words, A system is said to be in equilibrium when it manifests no changes with time.

For a system to be in thermodynamic equilibrium, it must ^{attain} satisfy the 3 types of equilibrium.

1. Thermal equilibrium :- Equality of temperature.

The temperature of the system does not change with time and has same value at all points of the system e.g. If a system is separated from its surrounding by a diathermic wall (which allows heat to flow) and if there is no spontaneous change in the temperature of system, the system is said to be in thermal equilibrium.

2. Mechanical equilibrium :- There are no unbalanced forces within the system or between the surroundings and system. $\Rightarrow$ Equality of Forces.
The pressure in the system is same at all points and does not change w.r.t time.

3. Chemical equilibrium :- No chemical reaction takes place in the system and the chemical composition which is same throughout the system does not vary with time $\Rightarrow$ Equality of chemical potential.

When the conditions for any of the 3 types of equilibrium are not satisfied, a system is said to be in non-equilibrium state.

Process : Change of state of a system

Types of processes

Reversible

If a process comes back to its original state, without making any changes to the surroundings.

Follow the same path while doing so $\rightarrow$ No changes to the system or surroundings.

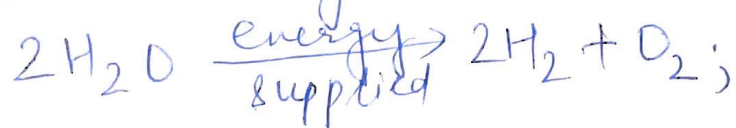
- Most ideal process. Takes place at slow rate.

Examples :

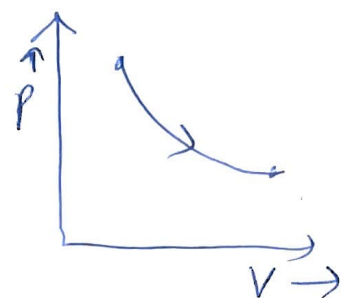
① Melting of ice $\sim 0^\circ\text{C}$.

Ice melts when a certain amount of heat is absorbed by it. Now, water formed can be converted into ice if same amount of heat is removed.

② Electrolytic Process



If reversed $\rightarrow$ adequate amount of 2H_2 and O_2 kept together, water formed with release of energy.



Irreversible

A process which cannot be reversed is called irreversible process.

- Can be carried out in one direction only.
- It occurs at a finite rate.

Examples -

① Frictional motion like a moving block sliding on a rough surface; ~~Working~~ is done while moving it and in bringing it back to original state, work has to be done again.

② Mixing of 2 different substances.
like milk into tea.

③ Rusting of iron

Conditions for Reversible processes

In order for a process to be reversible, a few conditions are necessary.

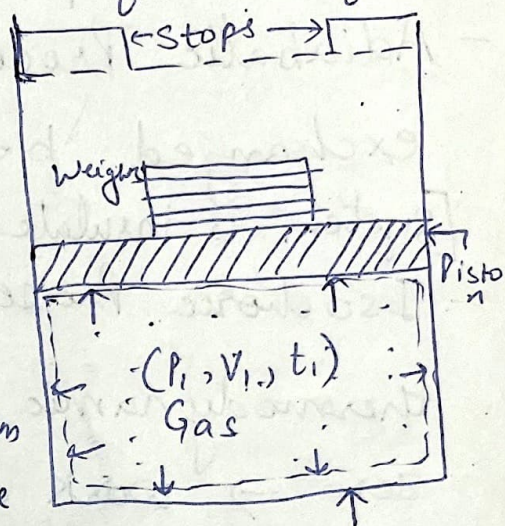
- There should be no heat or energy losses like friction / resistance etc.
- Change in pressure and volume should be at infinitesimally slow rate like in Quasi-Static processes.

Quasi-Static Process

Almost - Rest.

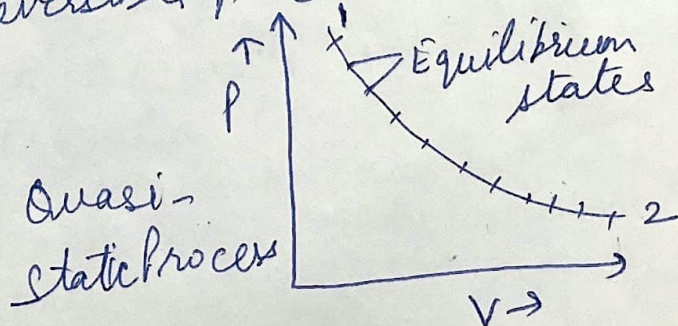
This process is a succession of equilibrium states and is carried out in a very slow manner. Infinite slowness is the characteristic feature of a quasi-static process.

Consider a system of gas in a cylinder in equilibrium state (P_1, V_1, T_1) . The weight on piston just balances the upward force exerted by the gas. If all weights removed together, piston will move to stops and new equilibrium (P_2, V_2, T_2) will be reached. But the



Intermediate states would be non equilibrium states. Now if the weights (small ones) are removed one by one slowly, the departure of the system from the thermodynamic equilibrium state will be infinitesimally small. So every state passed will be an equilibrium state. So this is a quasi-static process.

If the same small pieces of weights are now placed back slowly on piston, piston will move down slowly from state 2 to 1 following the same quasi-static path of succession of equilibrium states. So a quasi-static process is also a reversible process.



Types of Thermodynamic Processes.

- **Isothermal Process** :- In which the temperature remains constant. [A change in state of the system at constt. temperature].
- **Adiabatic Process** :- In which no heat is exchanged between the system and surroundings. [System is insulated from its env., state changes but no heat transfer].
- **Isochoric Process** :- Change in volume of thermodynamic system is zero. Volume change zero $\Rightarrow$ work done is zero.
- **Iso baric Process** :- Pressure remains constant during this process.

Microscopic & Macroscopic approach → Thermodynamic classification

2 diff. approaches / point of view from which behaviour of matter can be studied.

Macroscopic approach

Certain quantity of matter is considered without taking into account the events occurring at molecular level. Also known as classical thermodynamics.

We take "averaged" information in this. ⇒ values of properties (like P , V , T) are their average values.

↳ simpler approach

Microscopic

Study at molecular level. Each molecule has a certain position, vel., energy etc. which changes frequently. ⇒ Behavior of individual molecules is summed up using statistical methods to know the overall behaviour of system. Also called statistical thermodynamics.

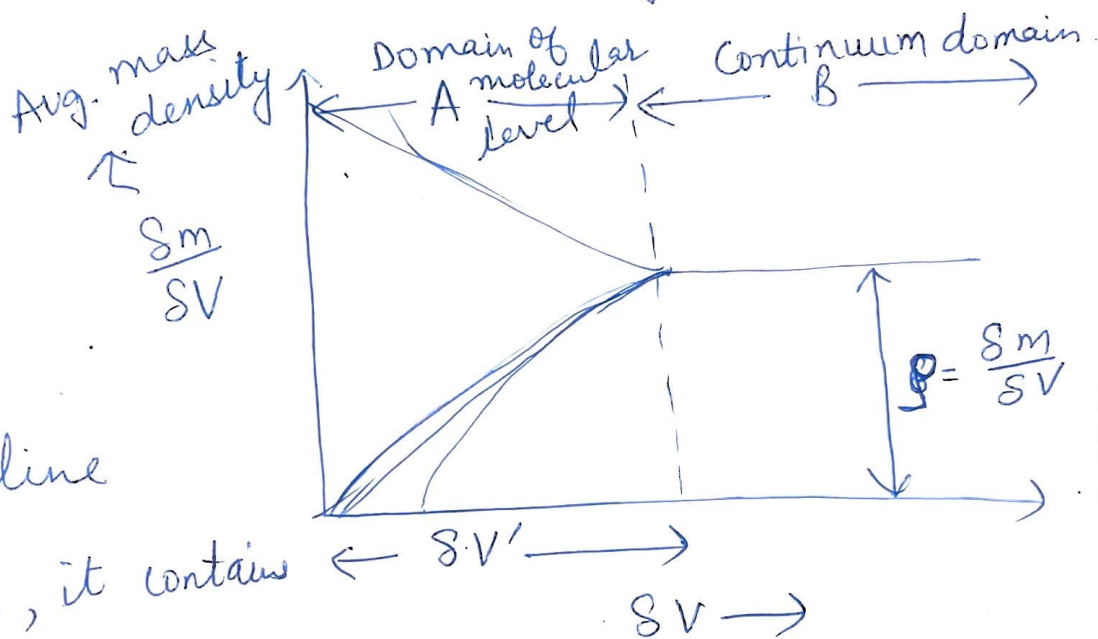
In our course, we will use the macroscopic approach, where a certain amount of matter or vol. is required. To decide, how much vol./matter, we use concept of continuum.

Reasons for using macroscopic approach

- ① 1 mm^3 of air → 10^{16} molecules. → Too much information to manage.
- ② Microscopic positions and velocities not useful for determining how macroscopic systems will act unless their effect is integrated.

Concept of Continuum : Valid in macroscopic approach.
volumes should be very large compared to molecular dimensions.

→ Gives the minimum volume to consider the system as continuous and apply "averaging" approach.



$SV \rightarrow$ large

$\frac{Sm}{SV}$ is a line

As SV reduces, it contains fewer molecules.
When molecules are less and vol. is small, avg. density fluctuates with time as molecules pass in and out.

Read this graph from right to left.

No def. avg. density.
Concept of continuum invalid.

av. ~~ge~~ density $\rho = \lim_{SV \rightarrow SV'} \frac{Sm}{SV}$

① Path → Avg distance after 2 collisions
③ ← ②

liquids and solids → concept of continuum valid always

Gases → Randomness → Not always valid. → Knudsen no.
 $Kn = \frac{\text{mean free path of molecule}}{\text{Dimension of vessel}}$; $Kn < 0.01 \rightarrow$ valid
 $Kn > 0.01 \rightarrow$ invalid

Zeroth Law of Thermodynamics

The property which distinguishes thermodynamics from other sciences is temperature.

The temperature is a thermal state of a body which distinguishes a hot body from a cold body.

When two bodies at different temperatures are brought into contact, after some time they attain a common temperature and are then said to exist in thermal equilibrium.

Law: If two systems (bodies) are in thermal equilibrium with a third body ^(or individually) separately, then they are also in thermal equilibrium with each other.

eg. body A and C are in th. eq. with body B.

$$\begin{array}{|c|c|} \hline A & B \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline C & B \\ \hline \end{array} \Rightarrow T_A = T_C \quad \text{Even without contact.}$$

$\Rightarrow T_A = T_B \quad \Rightarrow T_C = T_B$

This is known as zeroth law of thermodynamics. This law was first formulated by R.H. Fowler in 1931. The first and 2nd law of thermo. already existed much before 1931. So it was named as zeroth law of thermodynamics since it should have preceded the first and 2nd laws to form a logical sequence.

The zeroth law of thermodynamics provides the basis for the measurement of temperature.

The temperature of a body is a property that determines whether or not a system is in thermal equilibrium with other systems.

The value of temperature can be measured ^{easily} by indirect measurement with calibrated instruments.

It is determined by bringing a second body, thermometer in contact with the system and allowing thermal equilibrium to be reached.

The value of temperature is found by measuring some temperature dependent property of the thermometer. Any such property is called a thermometric property. e.g. length for mercury-in-glass gas type of thermometer.

[In case of digital thermometers, thermistors or resistance temperature detectors (RTDs) are used to measure temperature. A thermistor is a type of resistor whose resistance varies significantly with temperature. RTDs are made of pure metals (like platinum) whose resistance changes in a predictable way with temperature].

Point Function and Path function

Thermodynamic properties are point functions, since for a given state, there is a definite value for each property.

When 2 properties locate a point on the graph (co-ordinate axes) then those properties are called as point functions. e.g. Pressure, Temp., vol. etc

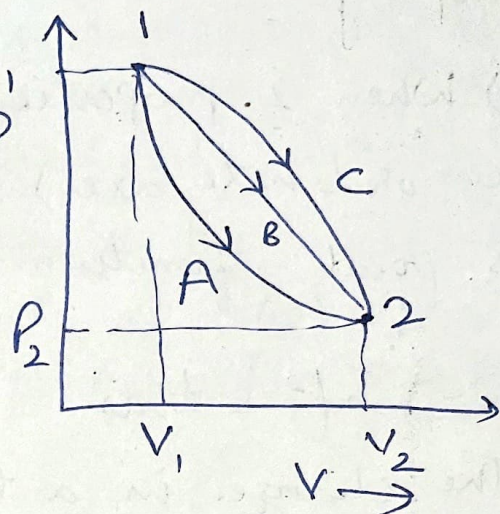
The change in a thermodynamic property of a system in a change of state is independent of the path the system follows during the change of state, and depends only on the initial and final states of the system. The differentials of point functions are exact or perfect differentials, and the integration is simply

$$\int_{V_1}^{V_2} dV = V_2 - V_1$$

The change in volume depends only on the end states of the system irrespective of the path the system follows.

Path function :- There are certain quantities which cannot be located on a graph by a point but are given by the area on that graph (area under the curve). Such quantities are called path function.

e.g. It is possible to take a system P , from state 1 to state 2 along many quasi-static paths, such as A, B or C.



Since the area under each curve represents the work for each process, the amount of work involved in each case is not a function of end states of the process, and it depends on the path the system follows in going from state 1 to 2.

So work is called a path function.

Heat is also a path function.

δW is an inexact differential. Their changes cannot be written as diff. b/w their end states.

Thus, $\int_1^2 \delta Q \neq Q_2 - Q_1$ or $\int_1^2 \delta W \neq W_2 - W_1$

Rather

$$\int_1^2 \delta W = W_{1-2} \text{ or } W_2$$

δW or dW to diff. from dV/dP (exact diff.)

Heat and Work.

Heat $\rightarrow$ form of energy. Can be converted into mechanical work and vice versa. e.g. Heat produced by rubbing of hands. On the other hand, any device which converts heat energy into mechanical work is called a heat engine $\rightarrow$ Examples are Steam engine, diesel engine, petrol engine.

Joule's mechanical equivalent of heat $\leftarrow$ [Joule gave a relation b/w work and heat]
 $W \propto Q$
 $W = JQ$
Whenever work is transformed into heat or heat $\rightarrow$ work, quantity of work is mechanically equiv. to quantity of heat.

If $Q = 1$, $W = J$.

J is defined as amount of work done in producing unit quantity of heat. Units in SI = Joules for both W and Q or otherwise Q in calories, W in joules

$$J = 4.18 \text{ J/cal.}$$

Work Done By a Thermodynamic System (Closed system work)
 $dW = F \cdot dx$ $\leftarrow$ If a force F produces an infinitesimal displacement dx , the work done by the force is

External Work $\rightarrow$ force exerted by the system as a whole on its surroundings causing displacement e.g. gas expanding due to pressure and pushing piston out.

Internal Work $\rightarrow$ Work done by one part of the system on another part of same system e.g. in actual gas expands, work is also done against the molecular attractions b/w molecules.

Computing work done :- Gas in a cylinder

$$dW = F dx$$

$F \rightarrow$ Force exerted by gas on piston.

$dx \rightarrow$ displacement of piston by expansion of gas.

If A is the area of piston and P is the pressure at the instant when displacement dx is taking place.

$$dW = PA dx = PdV$$

$dV = A dx \rightarrow$ Volume's differential increase.

Total work done W by the gas on the piston

$$W = \int dW = \int_{V_i}^{V_f} P dV$$

Indicator Diagram :- P - V ~~curve~~ ^{graph}

Work done is equal to the area enclosed by the P - V diagram and volume axis.

$EF GH \rightarrow$ width dV at point E of P - V graph.
Strip $E \& F$ infinitesimally close.

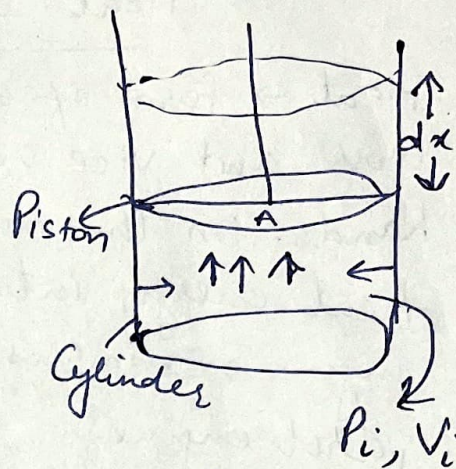
So $EF GH \rightarrow$ rectangle $\Rightarrow P$ at $E \& F$ equal work done

Area of this strip is $P dV$.

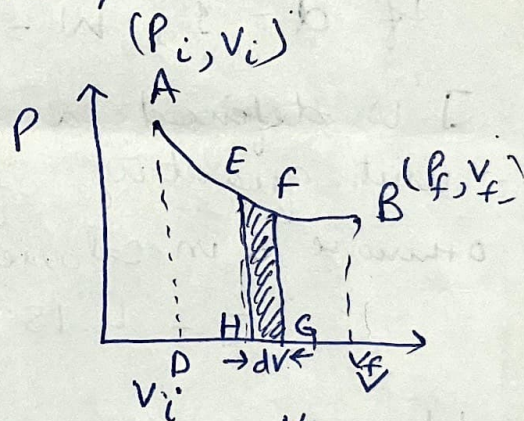
$P dV =$ Amount of work done in small change in volume dV .

Total work done $W =$ Sum of areas of infinitesimal narrow strips
 $= \sum P dV = \text{Area } ABCD.$

$$W = \int_{V_i}^{V_f} P dV = \text{Area of } PV\text{-graph}$$



reaches some equilibrium
 P_f, V_f



$$W = \int_{V_i}^{V_f} P dV$$

$\uparrow$ $\rightarrow$ If Work done by the system $\rightarrow$ vol. of gas increases $\rightarrow$
 Expansion $\rightarrow$ W is +ve
 If work done ~~on~~ the system $\rightarrow$ vol. of gas decreases $\rightarrow$
 compression $\rightarrow$ work done -ve

Work done in Cyclic Process.

$$W_{ABC} = \text{Area under } ABC$$

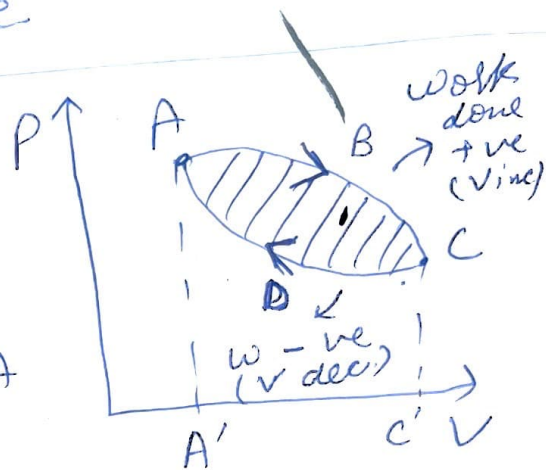
$$= ABC C' A' A$$

$$\text{Hky } W_{CDA} = \text{Area under } CDA$$

$$= ADC C' A' A$$

$$\text{Net work done} = W_{ABC} - W_{CDA}$$

$$= \text{Area } ABCDA \text{ (Shaded region)}$$



Work done in certain processes.

① Isothermal $\rightarrow$ T const. $\rightarrow$ Gas expanding isothermally.

$$dW = PdV \text{ (for a very small change)}$$

$$W = \int_{V_1}^{V_2} PdV$$

$$\rightarrow \text{As exp. isothermal } P = \frac{RT}{V}$$

$$PV = RT \text{ (one mole of gas)}$$

$$R \rightarrow \text{universal gas const. } 8.31 \text{ J/mole K}$$

$$\Rightarrow W = RT \int_{V_1}^{V_2} \frac{dV}{V} = RT \log_e \frac{V_2}{V_1} = 2.3026 RT \log_{10} \frac{V_2}{V_1}$$

② Adiabatic $\rightarrow$ No heat exchange $\rightarrow$ Gas expands in an insulated cylinder

$$W = \int_{V_1}^{V_2} PdV ; \text{ for an adiabatic process}$$

$$PV^\gamma = \text{const}$$

$$PV^\gamma = K$$

$$\Rightarrow P = \frac{K}{V^\gamma}$$

$$\therefore W = \int_{V_1}^{V_2} K \frac{dV}{V^r} = \int_{V_1}^{V_2} K V^{-r} dV = K \left[\frac{V^{-r+1}}{-r+1} \right]_{V_1}^{V_2}$$

$$= \frac{1}{1-r} \left[K V_2^{-r+1} - K V_1^{-r+1} \right]$$

$$= \frac{1}{1-r} \left[K V_2^{-r} V_2 - K V_1^{-r} V_1 \right]$$

As $PV^r = \text{const.} = K \Rightarrow P_1 = K V_1^{-r} \& P_2 = K V_2^{-r}$

$$\text{So } W = \frac{1}{1-r} (P_2 V_2 - P_1 V_1)$$

③ Isobaric Process : P const. during gas expansion.

$$dW = P dV \quad \text{or} \quad W = \int_{V_1}^{V_2} P dV = P (V_2 - V_1)$$

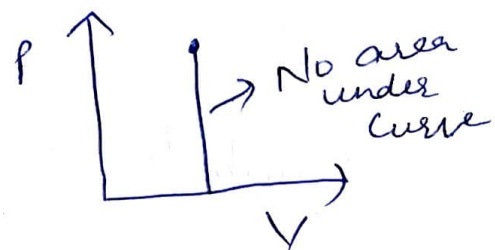
④ Isochoric Process : Volume const. during gas expansion.

$$dV = 0 \Rightarrow dW = 0 \quad \text{or} \quad W = \int_{V_1} P dV = 0$$

Numerical : Isothermal expansion,
Vol. of 1g mole of gas at 27°C is doubled
Calculate work done? $\rightarrow 300\text{K}$

$$W = 2.3026 RT \log_{10} \frac{V_2}{V_1} \rightarrow \log_{10} \frac{2V}{V} \downarrow \log_{10} 2$$

$$W = 2.3026 \times 8.3 \times 300 \times 0.3010 = 1725.8 \text{ J}$$



Comparison between Adiabatic and Isothermal curves $\div$ Or slope of adiabatic curve is γ times slope of isothermal curve.

Adiabatic

$$P V^{\gamma} = \text{const.}$$

Taking log

$$\log P + \gamma \log V = \log(\text{const.})$$

Differentiating

$$\frac{1}{P} \frac{dP}{dV} + \frac{\gamma}{V} = 0$$

$$\left(\frac{dP}{dV} \right)_{\text{ad.}} = -\frac{\gamma P}{V} \quad \text{--- (1)}$$

Isothermal

$$P V = R T = \text{const.}$$

$$\Rightarrow P V = \text{const.}$$

Taking log

$$\log P + \log V = \log(\text{const.})$$

Differentiating

$$\frac{1}{P} \frac{dP}{dV} + \frac{1}{V} = 0$$

$$\left(\frac{dP}{dV} \right)_{\text{iso}} = -\frac{P}{V} \quad \text{--- (2)}$$

Dividing (1) by (2)

$$\frac{\left(\frac{dP}{dV} \right)_{\text{ad.}}}{\left(\frac{dP}{dV} \right)_{\text{iso}}} = \frac{-\frac{\gamma P}{V}}{-\frac{P}{V}} = \gamma$$

$$\Rightarrow \frac{\text{slope adiabatic}}{\text{slope isothermal}} = \gamma$$

$$\text{or slope adiabatic} = \gamma (\text{slope isothermal})$$

Internal energy :- Energy possessed by a body or a system by virtue of its molecular arrangement and motion of molecules is called internal energy.

$U = U_K + U_P$ $U_K \rightarrow$ KE due to molecular motion
 $U_P \rightarrow$ PE due to intermolecular forces.

For ideal/perfect gases, internal energy U depends only on temp., molecular forces of attraction/repulsion assumed to be absent.

$$U = U(T) \quad \text{ideal gases.}$$

For ~~at~~ non permanent gas, vapour, or liquid, internal energy depends strongly on temp. and weakly on pressure.

$$U = U(T, P) \quad \text{for all gases/vapour/liquid}$$

Main points

- Internal energy is a point function and a thermodynamic property.
- Depends on mass $\Rightarrow$ Extensive property.

Heat

The energy transfer due to temperature difference is known as heat transfer.

$$Q \propto m$$

$$Q \propto \Delta t$$

$$Q \propto m \Delta t$$

$$\boxed{Q = mc \Delta t} \quad c = \text{Specific heat}$$

~~Specific~~ Specific heat is the amount of heat required to raise the temperature of unit mass of substance to unit degree temp. difference.

~~Main points~~

first law of thermodynamics

Energy Conservation. Heat is a form of energy.

"Energy can neither be created nor be destroyed it can only be transformed from one form to another."

$$Q = (U_2 - U_1) + W \Rightarrow \text{general form}$$

When a certain amount of ~~energy~~ heat is supplied to a system, it is spent in

- Internal energy $\rightarrow$ both in raising temp. of body $\hookrightarrow U_k$ and in doing internal work against molecular attractions which increases internal pot energy U_p . Total U is raised $= U_k + U_p$
- Remaining heat energy is spent in expanding the system i.e. in doing external work.

$$\Rightarrow \delta Q = dU + \delta W \quad \text{Differential form.}$$

$$\text{or } dQ = dU + dW$$

- For a closed system undergoing a cycle, net heat transfer is equal to net work transfer.

$$\sum Q_i = \sum W_i \quad \text{or } \oint \delta Q = \oint \delta W$$

Point func
Exact
diff $\Rightarrow$

$$\oint dU = 0$$

- For an isolated system

$$\delta Q = 0, \delta W = 0, U_2 = U_1 \Rightarrow dU = 0$$

Sign Conventions

1. Heat supplied to the system $dQ +ve$ (endothermic)
2. Heat released by the system $dQ -ve$ (exothermic)
3. Temp. increased - dU positive
4. Temp. decreased - $dU -ve$
5. Work done by the system $dW +ve$
6. Work done on the system $dW -ve$.

Note : First law of thermodynamics introduces the concept of Internal energy.

Heat Transfer in various processes

- 1) Isothermal process : Temp. ^{is kept} constant by adding heat or taking it away from the substance. As there is no change in temperature $dU = 0$
Acc to 1st law of thermodynamics

$$\delta Q = dU + \delta W \text{ as } dU = 0$$

$$\Rightarrow \delta Q = \delta W$$

$\therefore$ In an isothermal process -

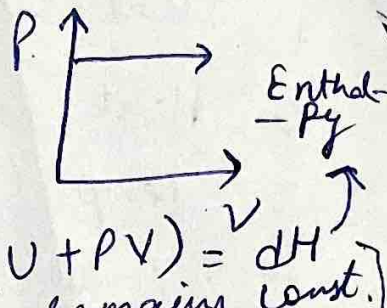
$$\begin{array}{l} \text{Heat added} \\ \text{(or released)} \end{array} = \begin{array}{l} \text{External work done by} \\ \text{the gas} \end{array} \quad \begin{array}{l} \text{by} \\ \text{(on the} \\ \text{gas)} \end{array}$$

(i) Iso baric process :- Pressure : const. , amount of heat supplied is used in doing external work and increasing volume dV .

$$\text{External work } \delta W = P dV.$$

Acc. to 1st law of thermody.

$$\delta Q / dQ = dU + P dV \text{ or } \delta Q = d(U + PV) = dH$$



(ii) Isochoric process :- Volume of system remains const. then $dV = 0$

$$dW = P dV = 0 \text{ as } dV = 0$$

Acc. to 1st law

$$dQ = dU + dW = dU$$

$$\Rightarrow \boxed{dQ = dU}$$

$$\begin{aligned} \text{Enthalpy} &= U + PV \\ \text{Extensive prop.} \\ \text{Specific Enthalpy} &= \frac{H}{m} = h, \\ \text{Intensive prop.} \end{aligned}$$

$\Rightarrow$ whole heat is used in increasing the internal energy only.

(iv) Adiabatic process :- No heat transfer $\Rightarrow$ No heat enters or leaves the system. Temp. does not remain const. though

Acc. to 1st law of thermodynamics

$$\delta Q = dU + \delta W$$

Since no heat transfer $\delta Q = 0$

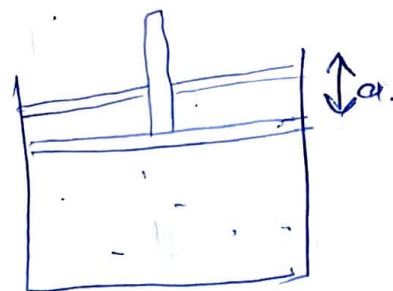
$$dU = \pm \delta W$$

$\Rightarrow$ When a gas is allowed to compress/expand, it work is done on it/by it and its energy gained appears as increase/decrease in internal energy.

Adiabatic eqn. of state derivation :-

Work done by piston

$$\delta W = f \cdot dx = P A dx = P dV$$



1st law, $\delta Q = dU + \delta W$. Adiabatic $\delta Q = 0$

$$\Rightarrow dU + P dV = 0 \quad (1)$$

specific heat $C_V = \frac{dU}{dT}$

$$\text{or } dU = C_V dT \quad (2)$$

$$C_V = \frac{\delta Q}{dT} \left(\begin{array}{l} \delta Q = C_V dT \\ \text{but at const vol.} \\ \delta Q = dU \end{array} \right)$$

Put (2) in (1)

$$C_V dT + P dV = 0 \quad (3)$$

Now, $PV = RT$; on diff. (total diff.)

$$P dV + V dP = R dT$$

$$\Rightarrow dT = \frac{P dV + V dP}{R} \quad (4)$$

Substituting (4) in (3)

$$C_V \left(\frac{P dV + V dP}{R} \right) + P dV = 0$$

$$\text{or } C_V P dV + C_V V dP + R P dV = 0$$

$$\text{or } (C_V + R) P dV + C_V V dP = 0$$

$$C_P P dV + C_V V dP = 0 \quad (5)$$

Dividing (5) by $C_V P V$

$$\frac{C_P P dV}{C_V P V} + \frac{C_V V dP}{C_V P V} = 0$$

$$\Rightarrow \frac{C_P}{C_V} \frac{dV}{V} + \frac{dP}{P} = 0 \quad \text{or} \quad \gamma \frac{dV}{V} + \frac{dP}{P} = 0 \quad (6)$$

Integrating (6) $\rightarrow \gamma \log V + \log P = \text{const.}$

$$\log P V^\gamma = \text{const.} \Rightarrow P V^\gamma = \text{const.} = K$$

$$\Rightarrow \boxed{P_1 V_1^\gamma = P_2 V_2^\gamma}$$

Case I

$$P V^{\gamma} = \text{const.}$$

$$P V = R T \Rightarrow P = \frac{R T}{V}$$

$$\Rightarrow \left(\frac{R T}{V} \right) V^{\gamma} = \text{const.}$$

$$R T V^{\gamma-1} = \text{const.}$$

$$\text{or } T V^{\gamma-1} = \text{const.} / R = \text{const.}$$

Case II

$$P V^{\gamma} = \text{const.}$$

$$P V = R T \Rightarrow V = \frac{R T}{P}$$

$$\Rightarrow P \left(\frac{R T}{P} \right)^{\gamma} = \text{const.}$$

$$\frac{P T^{\gamma}}{P^{\gamma}} = \frac{\text{const}}{R} = \text{const.}$$

$$T^{\gamma} P^{1-\gamma} = \text{const.}$$

$$\text{or } P^{\gamma-1} T^{-\gamma} = \text{const.}$$

Limitations of first law of thermodynamics

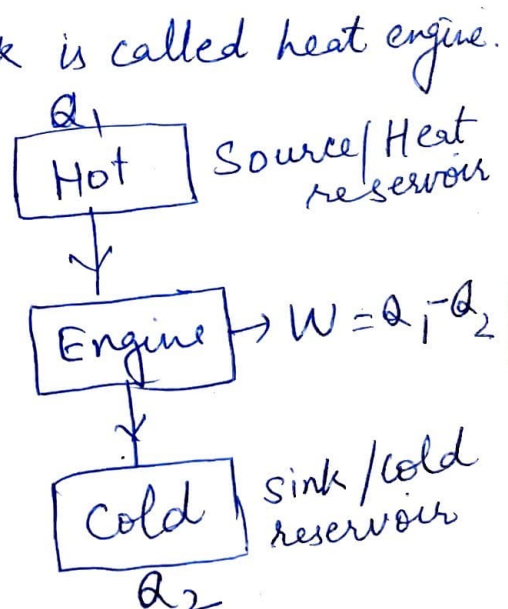
- ① Considers all forms of energy equivalent $\rightarrow$ measuring molecular energy challenging
- ② Does not consider direction of energy equivalence.
- ③ Doesn't address efficiency of energy conversion.
- ④ Does not account for entropy.

Heat Engine

Machine that can convert heat into work is called heat engine.

Heat engine must operate continuously, it absorbs heat at higher temperature and rejects it at lower temperature.

The material used in the operation of engine is called working substance.
e.g. in steam engines, working substance is steam.



Efficiency η

$$\eta = \frac{\text{Net work done}}{\text{Heat absorbed}} = \frac{W}{Q_1}$$

For one complete cycle.

After one cycle, engine returns to its initial state. So $\Delta U = 0$
 $\Rightarrow$ Change in internal energy = 0

From first law $Q_1 - Q_2 = \Delta U + W$

As $\Delta U = 0$ for ~~only~~ cycle, $\boxed{W = Q_1 - Q_2}$ (2)

$Q_1 \rightarrow$ Heat absorbed, $Q_2 \rightarrow$ Heat released,
 $W \rightarrow$ work done.

From (1) and (2)

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$\Rightarrow \boxed{\eta = 1 - \frac{Q_2}{Q_1}} \rightarrow \eta$ is always less than unity.

So efficiency is always less than 100%.

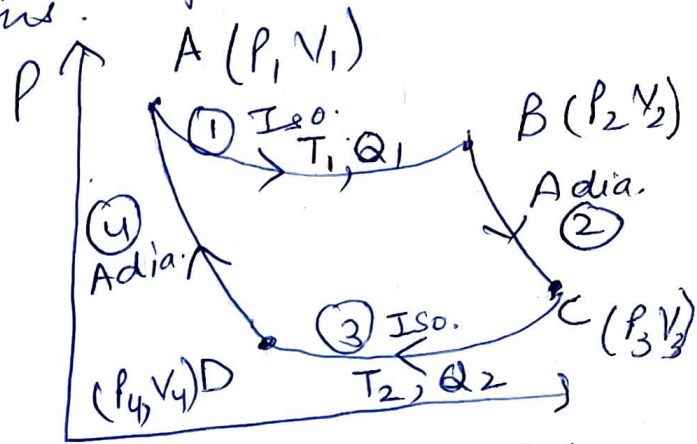
Dr. Nishu.

→ Completely reversible (piston assumed to be frictionless).

Carnot Cycle → Ideal heat engine → impossible to achieve in real situations.

4 stages :-

- ① Isothermal expansion
- ② Adiabatic expansion
- ③ Isothermal compression
- ④ Adiabatic compression



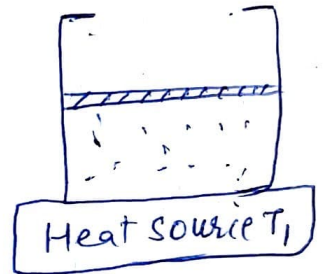
① Isothermal Expansion :- AB on graph (temp T_1 const) V

Gas in cylinder goes from P_1, V_1 to P_2, V_2 .
at const. Temp. T_1 → Substance/gas absorb heat Q_1 from the source at T_1 .

$$Q_1 = W_1 + (U_2 - U_1) \text{ or } \delta Q_1 = dU + \delta W_1$$

But $dU = 0$ as T const.

$$\Rightarrow Q_1 = W_1 = \int_{V_1}^{V_2} P dV = RT_1 \log_e \frac{V_2}{V_1}$$



② Adiabatic Expansion :- (B-C on graph) Cylinder removed from source and kept on insulating stand and undergo reversible adiabatic expansion. → No heat exchange
Temp. falls from T_1 to T_2 . [$P_2, V_2 \rightarrow P_3, V_3$]

$$W_2 = \frac{1}{1-\gamma} (P_3 V_3 - P_2 V_2)$$

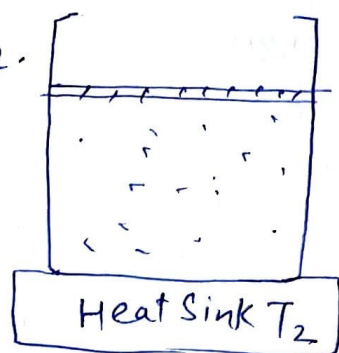
$$= \frac{1}{1-\gamma} (RT_2 - RT_1) \text{ or } \frac{R(T_1 - T_2)}{\gamma - 1}$$

$$\begin{bmatrix} PV = RT \\ P_3 V_3 = RT_2 \\ P_2 V_2 = RT_1 \end{bmatrix}$$

as $T_1 > T_2$

Now, gas is at a low pressure after 2 expansion processes. Now to complete a cycle; gas is compressed in 2 stages.

③ Isothermal Compression $\div$ C-D on graph (Temp T_2 constant)
 Cylinder removed from insulating stand and placed on sink at Temp T_2 .
 Piston moves inward (compression)
 Work done on the gas (-ve work).
 Heat produced in compression
 Q_2 is rejected to the sink at T_2 .
 (as isothermal $\rightarrow T$ constant).



$$Q_3 = W_3 = \int_{V_3}^{V_4} P dV = RT_2 \log_e \frac{V_4}{V_3}$$

As $V_4 < V_3$ (compression), $\Rightarrow W_3 = -RT_2 \log_e \frac{V_3}{V_4}$
 so Work -ve

④ Adiabatic Compression $\div$ D-A on graph ($T_2 \rightarrow T_1$)
 $P_4, V_4 \rightarrow P_1, V_1$

Cylinder removed from sink.

Again placed on insulating stand.

Gas compressed adiabatically and temp. rises.

Process continues until it goes back to its initial state. Full cycle completed.

$$W_4 = \int_{V_4}^{V_1} P \cdot dV \text{ or } \frac{R(T_2 - T_1)}{\gamma - 1} \text{ or } -\frac{R(T_1 - T_2)}{\gamma - 1}$$

-ve as work done on the system.

Net Effect of the Cycle

Total work done $W = W_1 + W_2 + W_3 + W_4$

$$= RT_1 \log_e \frac{V_2}{V_1} + \frac{R(T_1 - T_2)}{\gamma - 1} - RT_2 \log_e \frac{V_3}{V_4} - \frac{R(T_1 - T_2)}{\gamma - 1}$$

$$\Rightarrow W = RT_1 \log_e \frac{V_2}{V_1} - RT_2 \log_e \frac{V_3}{V_4}$$

Now, for B to C line on same adiabatic curve.

$$T_1 V_2^{r-1} = T_2 V_3^{r-1} \quad \text{or} \quad \frac{T_2}{T_1} = \left(\frac{V_2}{V_3} \right)^{r-1} \quad (1)$$

For A and D points

$$T_1 V_1^{r-1} = T_2 V_4^{r-1}$$

$$\left(\frac{T_2}{T_1} \right) = \left(\frac{V_1}{V_4} \right)^{r-1} \quad (2)$$

from (1) and (2)

$$\frac{V_1}{V_4} = \frac{V_2}{V_3} \Rightarrow \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Using in above eqn. for total W.

$$W = R(T_1 - T_2) \log \frac{V_2}{V_1}$$

Efficiency of Carnot engine

$$\eta = \frac{W}{Q_1} \quad \text{or} \quad \frac{W}{W_1}$$

as $Q_1 = W_1$ in 1st isothermal expansion

$$\Rightarrow \eta = \frac{R(T_1 - T_2) \log V_2/V_1}{R T_1 \log V_2/V_1} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \boxed{\eta = 1 - \frac{T_2}{T_1}}$$

Eff. of Carnot Engine

Just to Remind:

$$PV^r = \text{const.}$$

$$PV = RT \quad \text{or} \quad P = \frac{RT}{V}$$

$$\left(\frac{RT}{V} \right) V^r = \text{const}$$

$$\text{or } RT V^{r-1} = \text{const.}$$

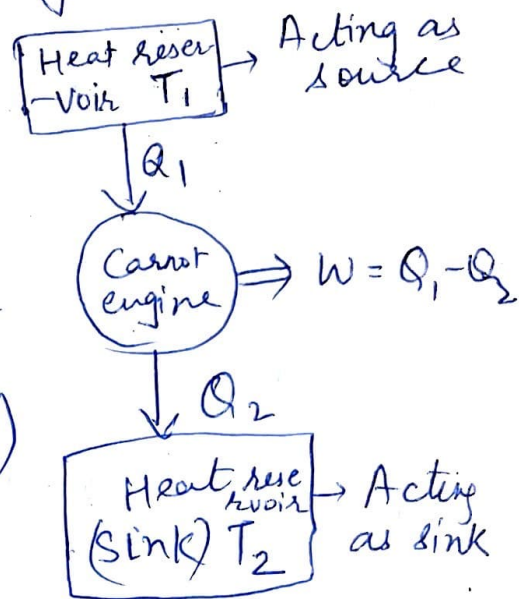
$$\text{or } T V^{r-1} = \frac{\text{const}}{R} = \text{const.}$$

$$\Rightarrow T_1 V_1^{r-1} = T_2 V_2^{r-1}$$

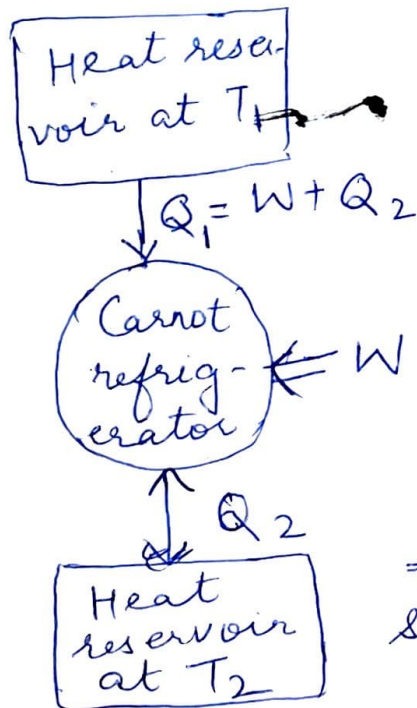
Carnot Engine and Carnot Refrigerator

Carnot cycle is perfectly reversible. It can work as a heat engine and also as a refrigerator.

- ① When it is operated as a heat engine, it absorbs heat Q_1 from source at temp. T_1 and does some work W and reject heat Q_2 to the sink at temp. T_2 ($T_2 < T_1$)
- [Converted Heat energy to Mechanical (work)]



- ② When operated as a refrigerator



When operated as a refrigerator, it absorbs heat Q_2 from the sink at temp. T_2 .

W amount of work is done by external means and reject heat Q_1 to the heat reservoir (source) at T_1 ($T_1 > T_2$).

→ In this case, work done on the system W and output is heat. Q_2 heat is removed from lower temp. reservoir.

C.O.P. (β)
Coefficient of Performance

$$= \frac{\text{Heat output}}{\text{Work done}} = \frac{Q_2}{Q_1 - Q_2}$$

$$\text{or } \frac{T_2}{T_1 - T_2}$$

(as for ideal Carnot cycle)
 $\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$

Second law of Thermodynamics

The first law of thermodynamics gives an exact relation between heat and work, but does not tell us the limitation and condition for this conversion (e.g. how much heat is converted into work and whether the transformation itself can take place or not).

Kelvin's Statement:- It is impossible to get a continuous supply of work by cooling a body to a temperature lower than that of the coldest of its surroundings.

Clausius Statement:- It is impossible for a self acting machine, unaided by any external agency to transfer heat from one body to another at a higher temperature. $\rightarrow$ Heat always flow from hot to cold body.

Both statements of second law are equivalent.

Violation of one automatically leads to violation of

Both statements imply that the natural processes other involve some form of inefficiency or energy dispersion. ~~and~~ No heat engine can be 100% efficient.

$\Rightarrow$ Second law of thermodynamics also states that the entropy of the universe is increasing continuously as a result of the spontaneous/natural processes.

processes which happen on their own
don't require additional energy/force

Entropy :- Measure of molecular disorder / degree of randomness. Greater the molecular disorder; more is the entropy.

When a body absorb or reject a small amount of heat energy at temp T ; change in entropy

$$dS = \frac{dQ}{T} \quad (\text{calories}/^{\circ}\text{C}) \text{ or } \text{J K}^{-1}$$

Change in entropy during adiabatic process.

$$S_2 - S_1 = \int_{S_1}^{S_2} dS \quad ; \quad dS = \frac{dQ}{dT} \quad \text{as } dQ = 0 \Rightarrow dS = 0$$

$\Rightarrow$ Entropy of a system remains constant during an adiabatic change.

In reversible process also, entropy of system remains constant. (as system, in equilibrium throughout and surroundings)

In all irreversible processes, entropy increases. As irreversible processes are continuously occurring in nature, entropy of universe is increasing.

$\Rightarrow$ Entropy either constant or increasing, never decreases. This is the basis of 2nd law of thermodynamics $\Rightarrow$ Entropy gives the direction in which thermodynamic process can take place.

↳ All thermodynamic processes take place in such a ~~way~~ direction that entropy remains constant or increases. No direction was given by 1st law.

→ Reversible engine

Carnot Theorem:- Carnot engine is the most efficient engine between two given reservoirs. (two temp.)

Entropy change during irreversible process:-

Eff. of irreversible cycle $\eta' = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$

Eff. of reversible cycle $\eta = 1 - \frac{T_2}{T_1}$

Acc. to Carnot theorem.

$$\eta' < \eta \Rightarrow 1 - \frac{Q_2}{Q_1} < 1 - \frac{T_2}{T_1}$$

or $\frac{Q_2}{Q_1} > \frac{T_2}{T_1}$ or $\frac{Q_2}{T_2} > \frac{Q_1}{T_1}$

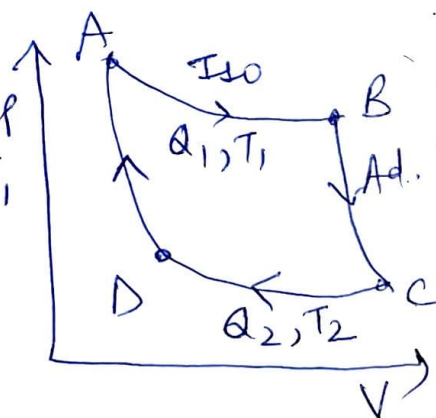
⇒ Entropy increases.

Entropy Change during reversible process:-

• Reversible Carnot cycle

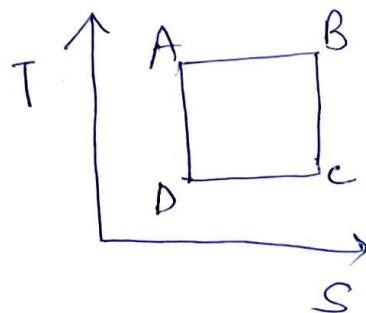
① Isothermal Expansion (A-B) - Q_1 heat absorbed at temp. T_1
temp. constant.

$$\int_A^B ds = \frac{Q_1}{T_1}$$



② Adiabatic Expansion (B-C)

$$\int_B^C ds = 0 \quad \text{as no heat transfer}$$



③ Isothermal Compression (C-D) : Reject heat Q_2 to the sink at temp. T_2

$$\int_C^D ds = -\frac{Q_2}{T_2}$$

④ Adiabatic Compression (D-A)

$$\int_D^A ds = 0$$

Net gain in entropy $\oint ds$

$$= \int_A^B ds + \int_B^C ds + \int_C^D ds + \int_D^A ds$$

$$= \frac{Q_1}{T_1} + 0 + \left(-\frac{Q_2}{T_2}\right) + 0 = \frac{Q_1}{T_1} - \frac{Q_2}{T_2}$$

For reversible cycle (Carnot cycle)

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \Rightarrow \oint ds = 0$$

$\Rightarrow$ Entropy unchanged in reversible cycle.